

timise the engine performance by having a more advanced software that runs on these control units and reduces fuel consumption.

Then there is connectivity. For example, when there is connectivity between ECUs, more information flows from navigation system into the power train, running the engine more optimally. Safety can also be improved with connectivity between navigation and driver assistance systems, as the information on an accident or other situation on the road can be used effectively in the car.

The connected vehicle is becoming a hot research topic, eventually leading to a lot of development. Connectivity not only helps to use information between the ECUs more effectively but also to communicate between vehicle-to-vehicle and vehicle-to-infrastructure. By having this information we can actually make a vehicle perform much better, both in terms of fuel consumption and safety.

Another trend is to have more standardisation on the ECUs because we think that it is possible to use and exchange or integrate software from multiple vendors. So there we have a standard called AutoSAR, which is basically an AUTomotive Open System ARchitecture. Bosch is a core member of AutoSAR consortium along with OEMs.

We have defined standards in Europe that are required to know how the software is structured on a vehicle. We feel that this will not only help integrate a lot of software coming from various vendors but will also enhance reuse of software so that we can focus on new functions.

Even though hardware platforms keep changing as microcontroller and memory chip costs go down following the Moore's law, the issue is always to re-engineer the software as soon as you change the hardware. To overcome that, AutoSAR helps transfer of software—independent of hardware architecture—from one ECU to another.

Hybrid vs electric—which side is the current technology favouring?

For pure electric technology, there is still a long way to go. Though we also do a lot of work on hybrid and electric systems, penetration depends quite a lot on the efficiency and cost of these systems. Basically, we still don't have cost-effective solutions for batteries, so storage of energy is a big issue.

We also work on electric vehicles that have conventional engines. We make sure that they run in the most optimal way by making use of the concept of storage in battery like brake recuperation. It means that while braking a vehicle, it is possible to feed the energy back, store it in a battery and then drive the vehicle using this restored energy. These sorts of systems are available only for high-end cars but now we are trying to develop systems that can also be used for low-priced vehicles.

What would be the future roadmap for a hybrid vehicle?

The simplest form of hybrid is 'start-stop.' The engine

can be switched off when the vehicle is stationary, and if it is possible to predict start and stop timings, power can be saved. The next level is to use it during braking, regenerate electric power from it, store it and then use it later. The highest level is pure electric and then a range extender. These are also projects where you switch on a generator only to charge the battery, otherwise the vehicle primarily runs on the battery.

What is the biggest challenge when working with mechatronics?

For mechatronic products, we need expertise not only from the mechanical or hydraulic side but also from the hardware and software side. Getting all these competencies together to develop products that will be very competitive in the market is a big challenge. On the experience level, we have been able to put a matrix team together that can work with all these competencies as very few people have all of them together. Assembling this team together was a big challenge, especially considering new technologies like Web 3.0.

Web 3.0 is a machine-to-machine communication, allowing autonomous management of systems. When talking about Web 3.0, I am not only talking about these multiple competencies but also knowledge of multiple systems. For example, there can be a car multimedia product with power train along with driver assistance and safety systems together adding value to the vehicle. So in that case, we need expertise from various domains coming together and then develop systems that can give value to the end customer who is driving the vehicle.

How is the Internet of Things (IoT) making its way to the automotive industry?

Use of Web 3.0 or the Internet of Things on the ECU gives the ability to connect vehicle-to-vehicle and vehicle-to-infrastructure and bring value to the driver. For example, if it is possible to predict the next signal light, or another vehicle around the corner in a U-turn, the safety of a vehicle can be increased.

In the future vehicle will become a part of the Internet. The vehicle itself becomes a transmitting unit of a lot of information in a car. For instance, it can be said that the car is like a weather station because temperature and other weather conditions of the particular area are known, which can be consolidated and used. This information could also be analysed for improvement of the engine performance or any other vehicle parameter.

The other way round is to take information from the outside into the car and provide it to the driver. Other trends include use of more number of cameras and various sensors like radar sensor. Radar sensor can detect the surrounding of a car and intelligently integrate information from the outside to make the car safer. ●

(The complete interview can be read at electronicsforu.com)